

ACME Operating Manual v4

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Revision History

Rev	Date	Change
4	2025-08-15	Added Section 6.4 on cross-references and updated Appendix A.
3	2025-01-10	Revised torque specifications in Section 4.2. See ECO-2025-003.
2	2024-06-01	Added Appendix A (Legacy Parts). Updated legal preamble.
1	2024-01-15	Initial release.

1. Introduction

This manual describes the installation, operation, and maintenance of the ACME Widget v4 platform. It supersedes all previous editions. For warranty terms, see Section 7.

2. Safety Precautions

Read all instructions before installation. Failure to follow these instructions may result in personal injury, property damage, or equipment failure. All electrical work must be performed by a qualified electrician in compliance with applicable codes.

3. Installation

3.1 Site Preparation

Ensure the mounting surface is level to within 2 mm per metre and capable of supporting the unit weight plus a 50% safety factor. Provide adequate ventilation (minimum 100 mm clearance on all sides). Install a dedicated circuit breaker per local codes.

3.2 Lubrication Procedure

Apply grease (NLGI Grade 2, Lithium complex) to all bearing surfaces before assembly. The bearing surfaces are identified by orange locating marks on the chassis. See section 4.2 for torque specifications. Torque all fasteners to the values listed in Table 4-1. Failure to follow this procedure may void warranty and cause premature bearing failure. Re-lubricate every 2,000 operating hours or annually, whichever occurs first.

4. Specifications

4.1 Component Specifications

Table 1 lists the key component specifications for the ACME Widget v4. All tolerances are manufacturing tolerances; refer to Section 4.2 for assembly torques.

Component	Part No.	Tolerance	Unit	Notes
Bearing A	PN-2041	+/-0.05	mm	Deep groove ball bearing
Bearing B	PN-2042	+/-0.05	mm	Cylindrical roller bearing
Shaft Seal	PN-3011	+/-0.02	mm	IP65 lip seal
Drive Gear	PN-4017	+/-0.10	mm	Module 2.5, 40 teeth
Motor Ctrl	PN-5001	N/A	N/A	24 V DC, 5 A max
Frame Bolt	PN-6001	+/-0.00	mm	M8 x 25, Grade 8.8

4.2 Assembly Torque Specifications

Torque specifications are provided in Newton-metres (Nm). Use a calibrated torque wrench. Apply Loctite 243 (medium strength) to all steel fasteners unless otherwise noted. Do not use impact tools for final torque.

Fastener	Size	Torque (Nm)	Threadlock
Frame mounting bolt	M8	25	Loctite 243
Bearing retainer	M5	8	None
Cover plate screw	M4	3.5	None
Drive gear nut	M12	70	Loctite 270
Terminal block	M3	1.2	None

5. Maintenance

5.1 Inspection Schedule

Perform the following inspections at the intervals shown. Record all inspection results in the maintenance log. Contact the service centre if any out-of-tolerance condition is found. Refer to the spare parts list in Appendix A for replacement part numbers.

Appendix A - Legacy Parts (scanned original, 1987)

APPENDIX A - LEGACY PARTS LIST

Part No.	Description	Stock	Unit
LG-0041	Drive Belt Assembly	12	EA
LG-0042	Bearing Set (4 per kit)	8	KT
LG-0043	Seal Kit (input shaft)	5	KT
LG-0044	O-Ring Set (assorted)	20	KT
LG-0045	Gasket Material Sheet	3	EA
LG-0046	Carbon Brush Set	10	PR
LG-0047	Contact Spring Assembly	6	EA

Note: Parts marked * are no longer manufactured.
Contact procurement for approved substitutes.

Last inventory count: 1987-03-01